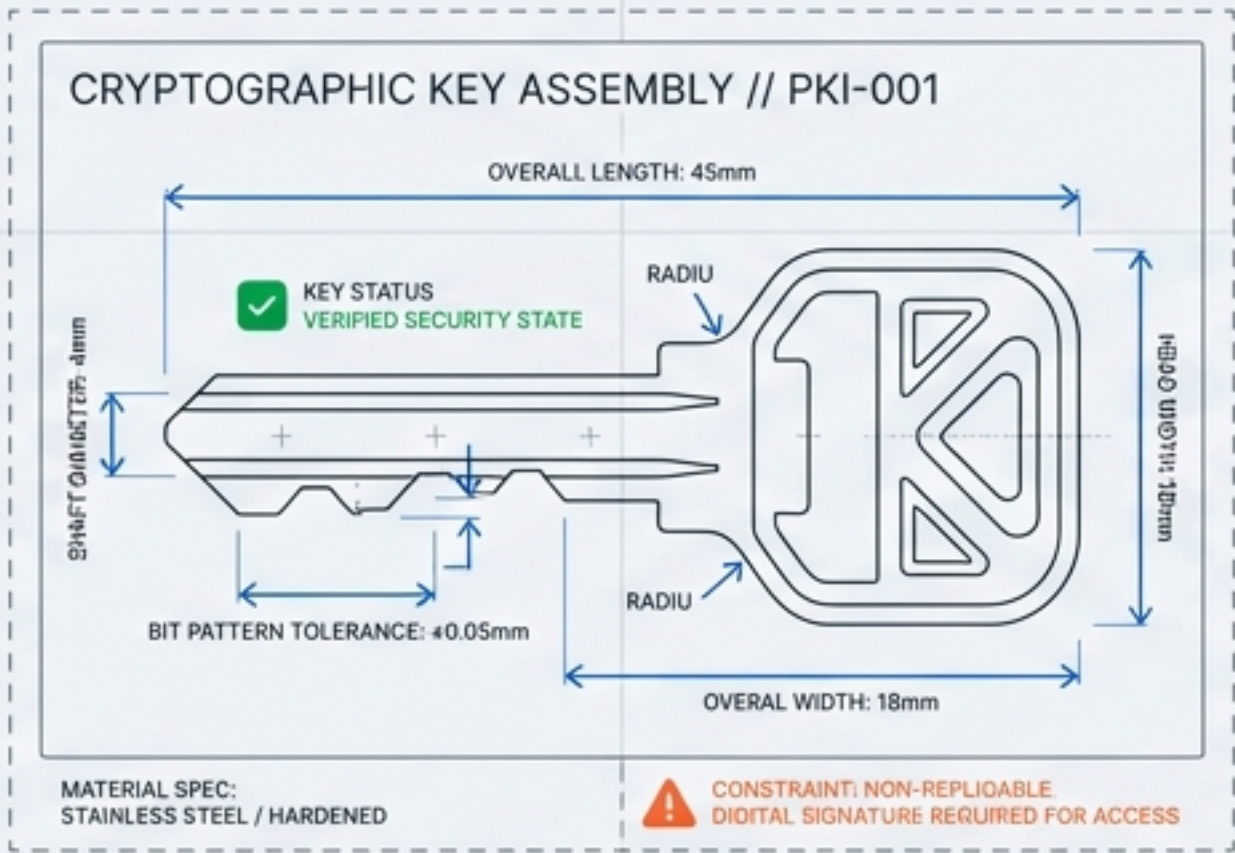


Admin PKI Key Management Page

Build Specification & Technical Architecture



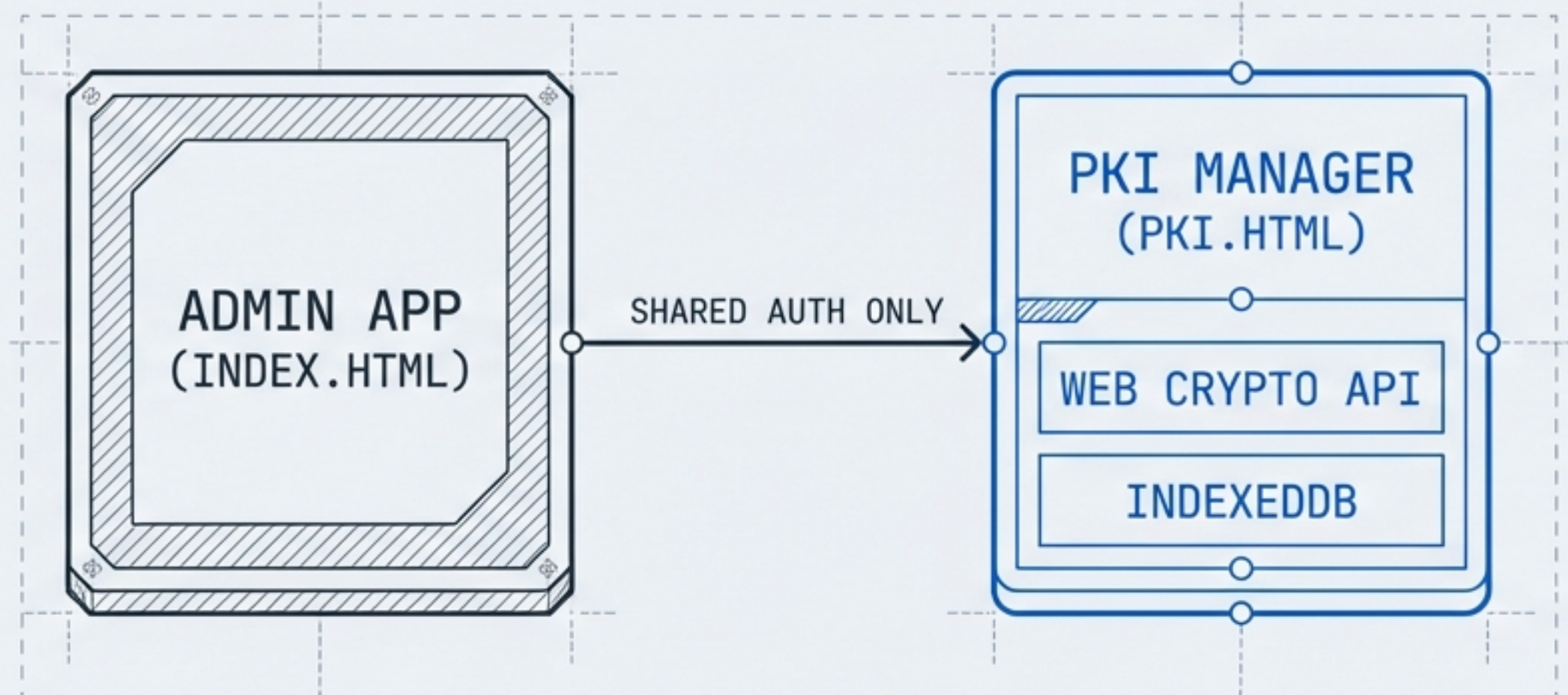
DATE:	19 Feb 2026
CLASSIFICATION:	B3 as B1 / D1
FROM:	Human (Project Lead)
TO:	Developer (Lead), Designer
TYPE:	Build Spec

THE MANDATE: A STANDALONE SECURE ENVIRONMENT

CORE DIRECTIVE: BUILD
A SEPARATE HTML FILE
FOR MANAGING WEB
CRYPTO KEY PAIRS.

LOCATION:
/ADMIN/PKI.HTML

DEPENDENCIES:
NATIVE BROWSER APIS
ONLY.

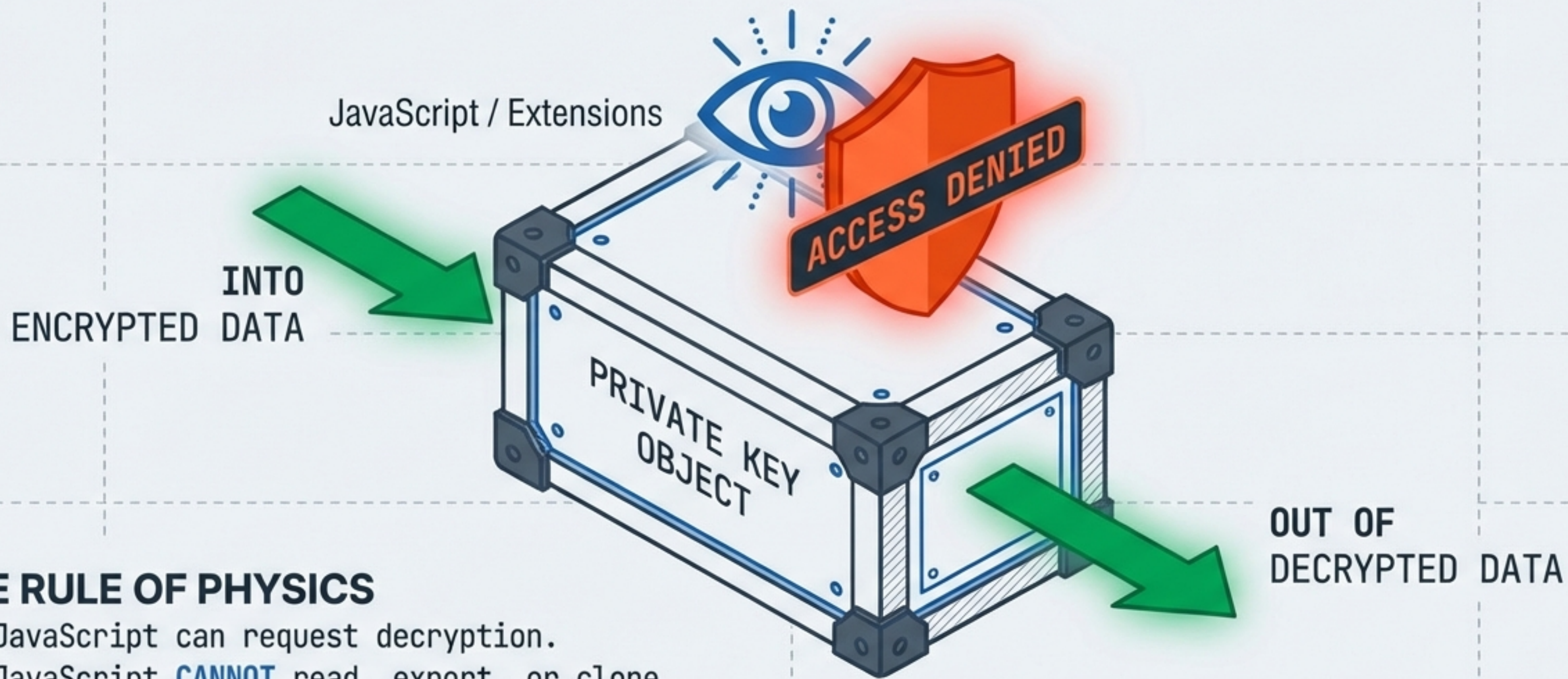


⚠️ CONSTRAINTS



FORBIDDEN: EXTERNAL LIBRARIES,
POLYFILLS, WASM.

THE SECURITY CORNERSTONE: `extractable:false`

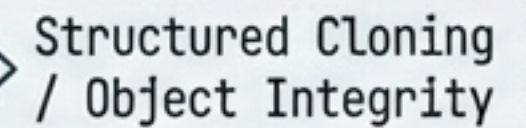


THE RULE OF PHYSICS

1. JavaScript can request decryption.
2. JavaScript **CANNOT** read, export, or clone the key material.
3. Key is strictly bound to the origin.

[illegible]

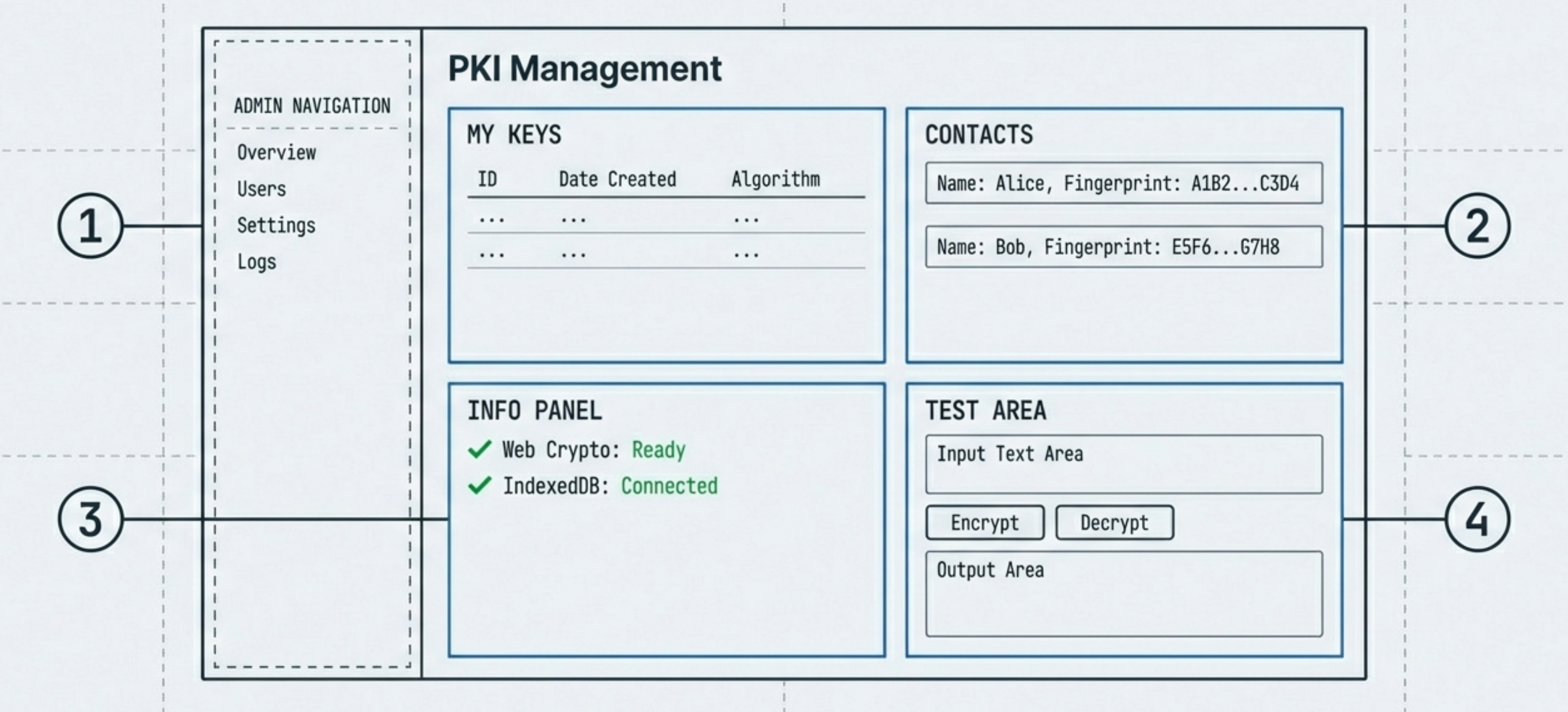
INDEXEDDB



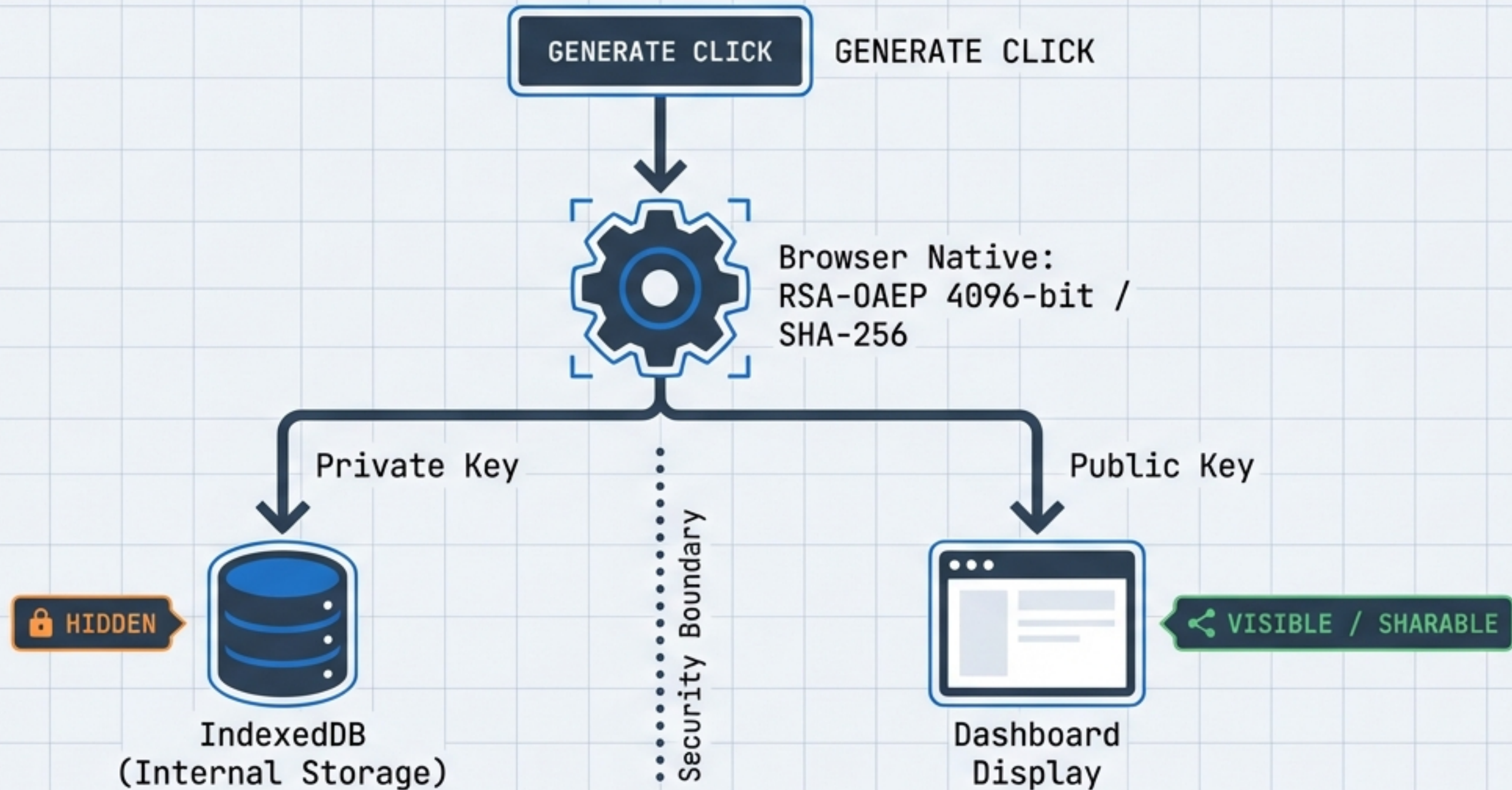
- Requires serialization (String conversion).
- Breaks "non-extractable" security flag.
- Exposure to XSS.

- Supports Structured Cloning.
- Preserves CryptoKey object integrity.
- Persists across browser restarts.

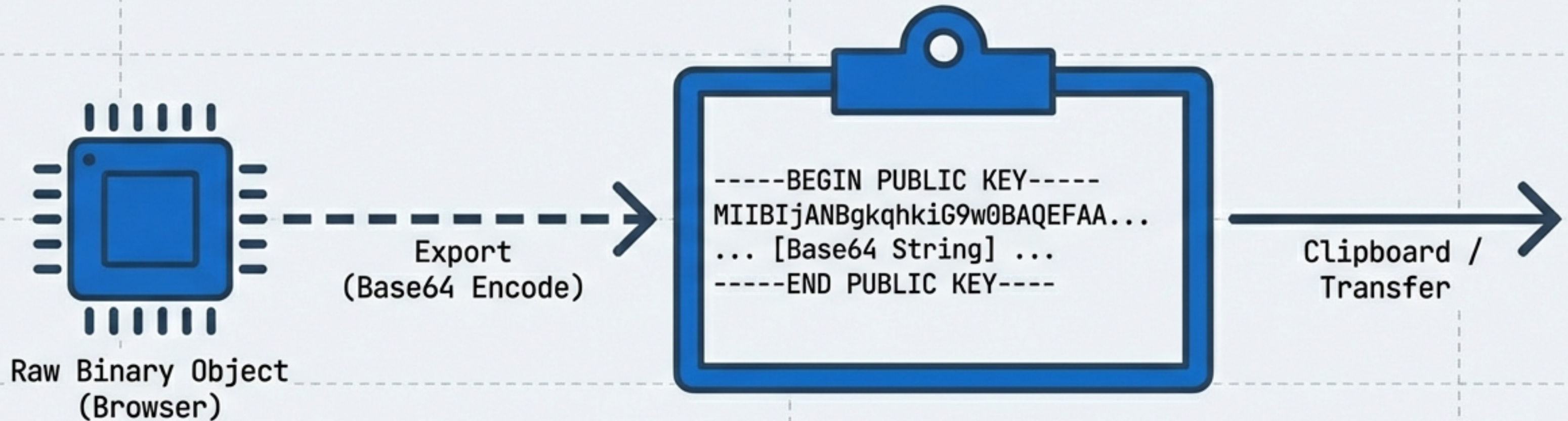
UI Geography: The Dashboard Wireframe



Workflow: Key Generation (RSA-OAEP)

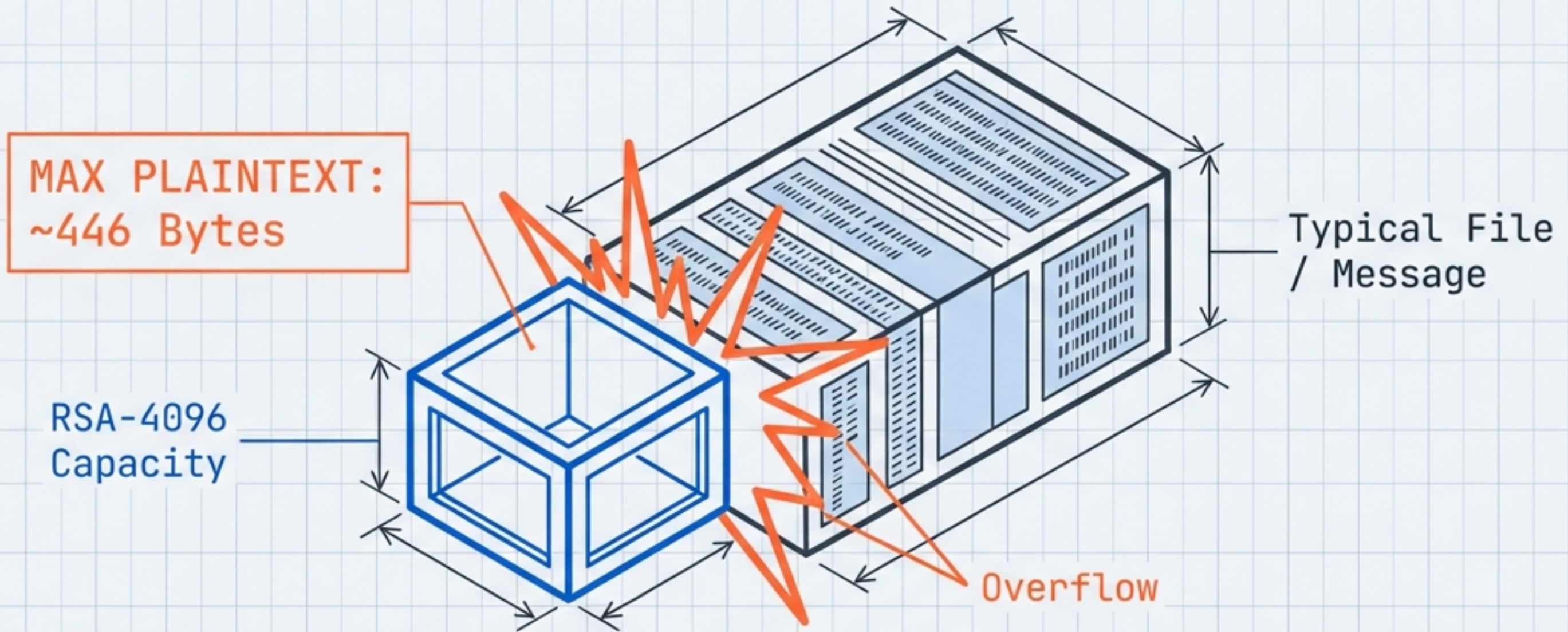


Interoperability: The PEM Standard



Public keys are extractable by definition.
This translation allows human sharing.

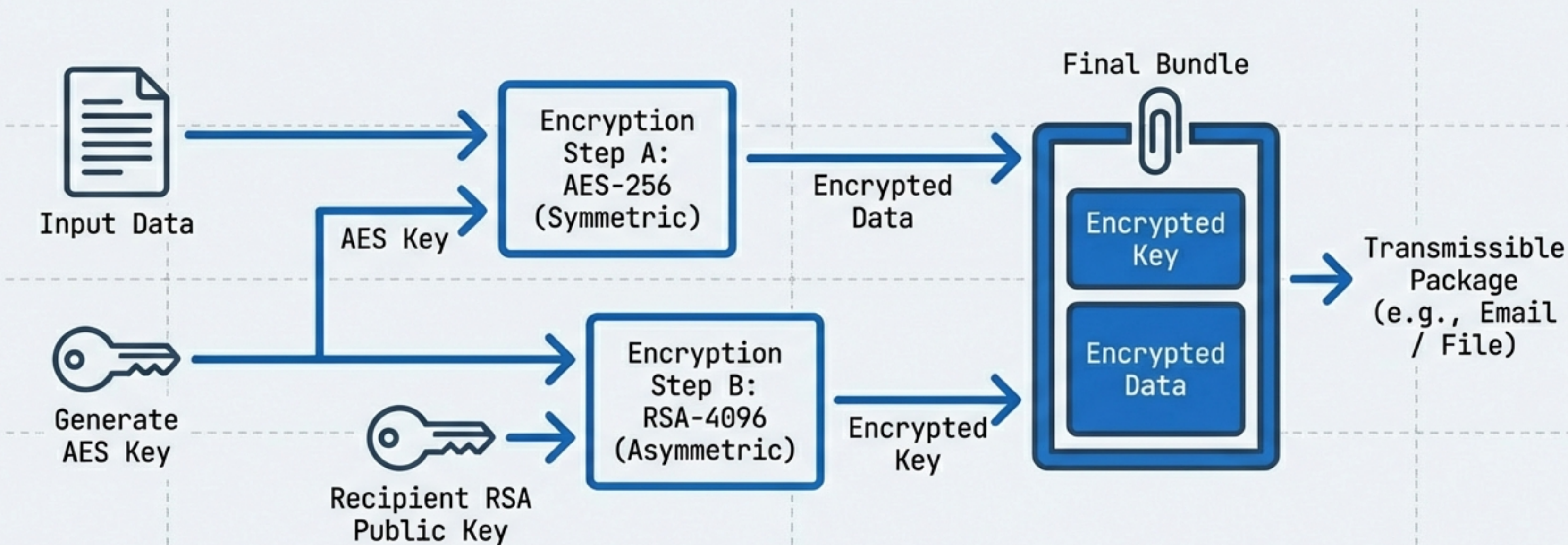
The Technical Constraint: RSA Size Limits



$$\text{Limit} = (\text{KeySize} / 8) - 2 * (\text{HashSize} / 8) - 2$$

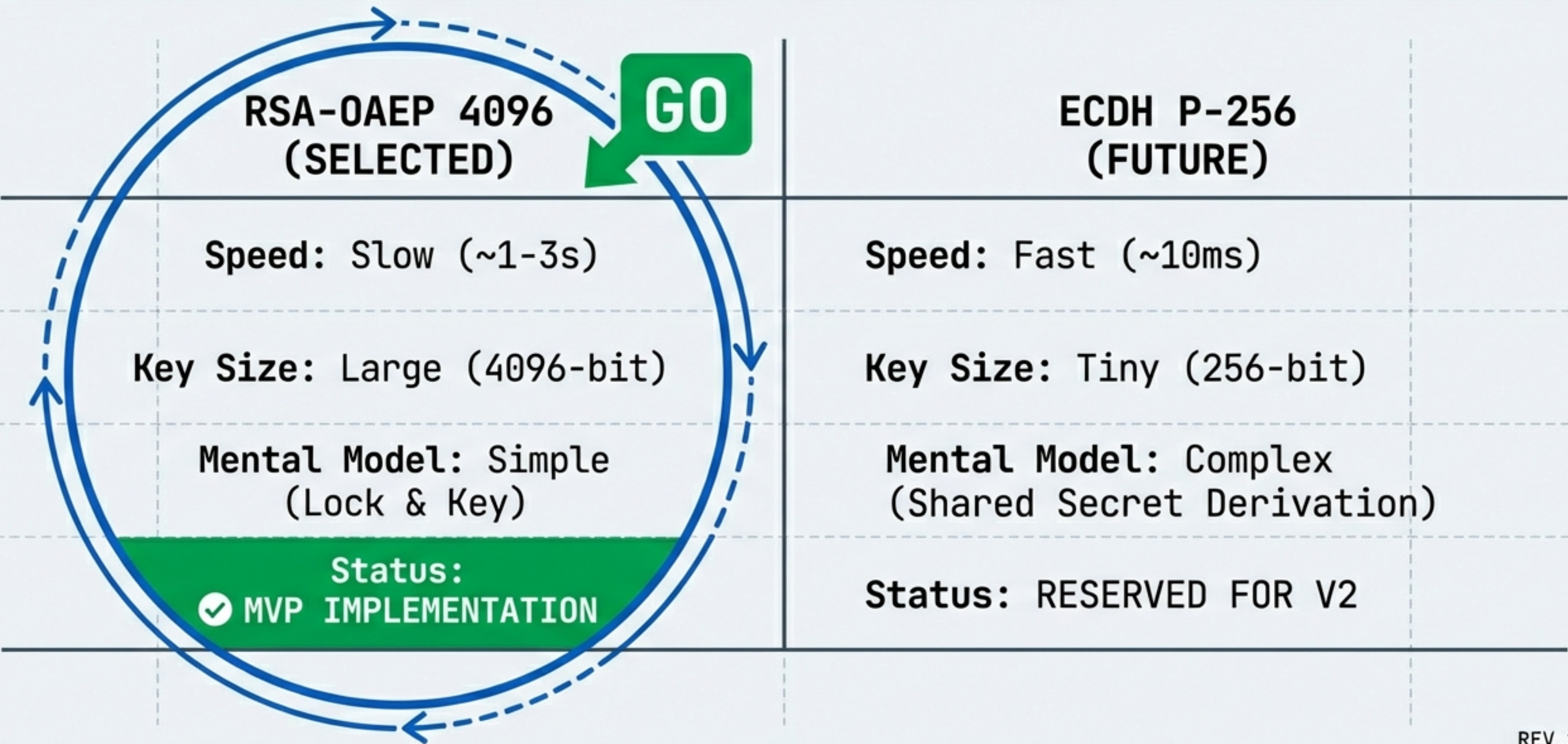
Direct RSA encryption fails for heavy data.

The Solution: Hybrid Encryption Architecture



Standard PGP / TLS Pattern

Algorithm Selection: Why RSA for MVP?



Browser Compatibility & Environment



v37+



v34+



v11+



v79+

Web Crypto API & IndexedDB Support



ZERO DEPENDENCY ARCHITECTURE:

NO Polyfills.

NO WASM.

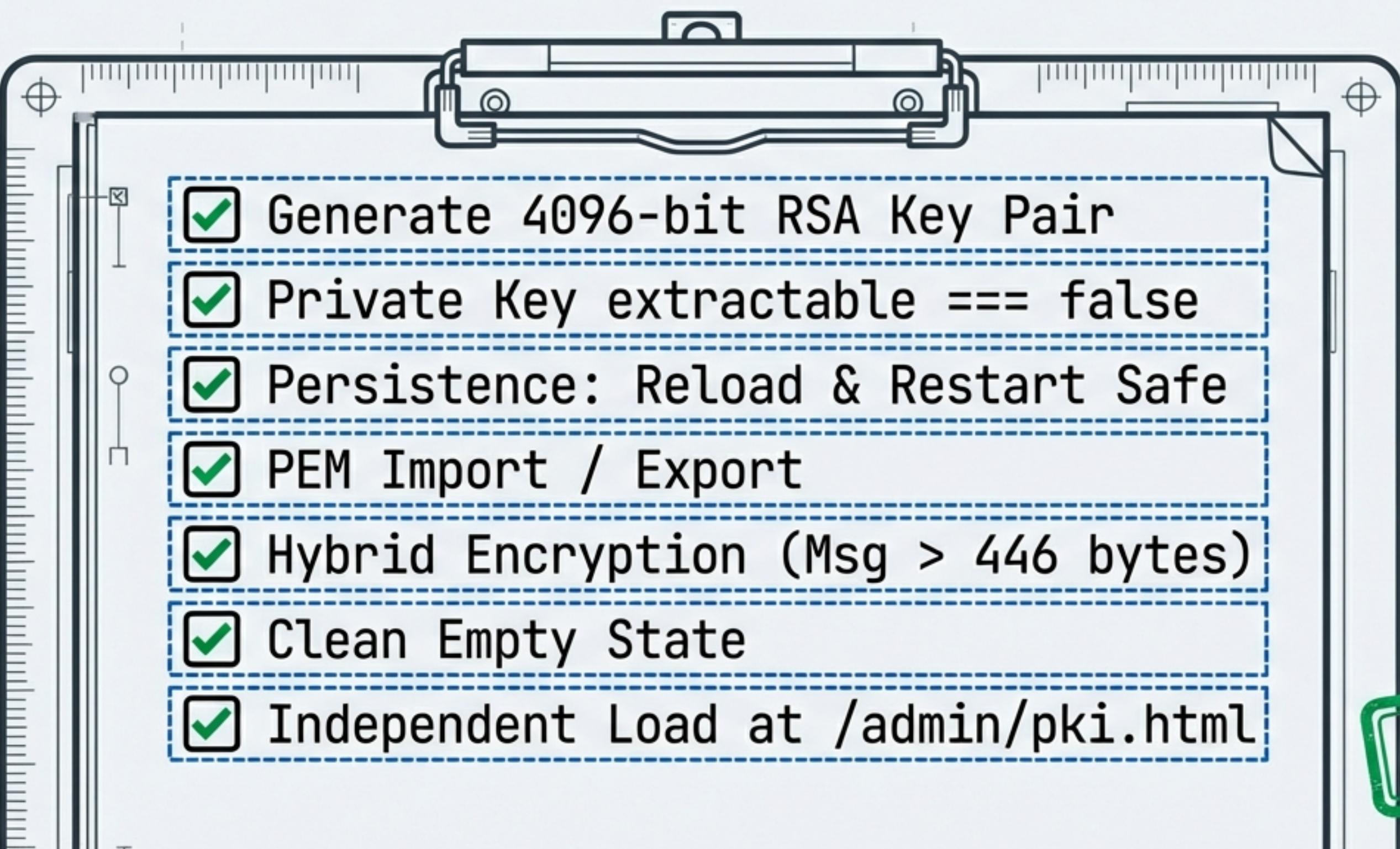
NO External Libraries.

Out of Scope (Do Not Build)

1. ~~Service Worker Trust Anchors~~
2. ~~Main App Integration (Send/Receive)~~
3. ~~Claude / LLM Integration~~
4. ~~Short Code Key Sharing~~

**CURRENT FOCUS:
Strict MVP Key Gen
+ Storage Only**

Acceptance Criteria: The Definition of Done

- 
- A clipboard with a checklist of acceptance criteria. The clipboard has a silver clip at the top and a ruler on the left side. The checklist is written on a piece of paper with a blue dashed border. Each item is preceded by a green checkmark in a box.
- ☒ Generate 4096-bit RSA Key Pair
 - ☒ Private Key extractable `=== false`
 - ☒ Persistence: Reload & Restart Safe
 - ☒ PEM Import / Export
 - ☒ Hybrid Encryption (Msg > 446 bytes)
 - ☒ Clean Empty State
 - ☒ Independent Load at `/admin/pki.html`

READY FOR DEV